Repeated Novel Geometries Are Addressable Only as Repeated Transitions

Ari Holtzman and NeuriCo

Abstract

Language models can learn new relational structure from context, but a learned internal geometry is useful to a user only if it can be addressed through ordinary prompts. We test this behavioral question for graph geometries inspired by recent work on in-context representation learning. We assign ordinary words to hidden nodes of either a 4x4 grid or a 16-node ring, expose the model to random-walk traces over the graph, and ask GPT-4.1 to answer bundled word-adjacency questions. We vary graph type, trace length, and prompt style, comparing direct JSON-only answers against an explicit-reasoning ablation and against an observed-edge memorization baseline. Across 1,280 scored query-level examples from 160 API calls, direct prompting reaches 72.7% accuracy and explicit reasoning reaches 73.9%, both below the 82.5% observed-edge baseline. The critical failure is sparse topology: on true adjacencies never observed in the prompt, direct prompting answers correctly in only 4/112 cases (3.6%), while explicit reasoning answers 0/112 correctly. The results show that these repeated novel geometries are verbally addressable mainly as memories of transitions already present in context. They do not support the stronger claim that ordinary prompting exposes unobserved parts of a newly induced topology under limited repetition.

1 Introduction

Can a prompt expose a geometry learned only in context? Language models are increasingly used as systems that define temporary worlds inside a prompt: names become entities, examples define relations, and users expect the model to answer questions about the induced structure. This expectation is stronger than next-token continuation. If a model internally organizes words around a new graph, a user still needs to know whether that graph can be queried in plain language.

Recent work on in-context representation learning gives this question a concrete form. Park et al. [2025] study graph tracing tasks in which familiar words are assigned to nodes of grids, rings, and related structures, and random-walk traces specify local transitions. They report that model representations can reorganize around the context-specified geometry as context grows. This is a striking representation-level result, but it leaves a behavioral gap: an internal geometry can exist without being addressable by an ordinary adjacency question.

Our main question is behavioral. We ask whether a current frontier language model can verbally address a newly specified graph geometry from random-walk context alone, especially when the relevant relation has not been repeated. The distinction matters because a high score on repeated edges can come from copying observed transitions rather than inferring the hidden topology. We therefore separate directly observed true edges from true edges that are sparse or absent in the trace.

We introduce GraphTrace-Address, a small but controlled prompt benchmark for this question. Each task assigns 16 ordinary words to either a 4x4 grid or a 16-node ring, samples a random-walk trace of length 16, 32, 64, 128, or 256, and asks eight bundled adjacency questions of the form `current=<word>, candidate=<word>`. We test GPT-4.1 with a direct JSON-only prompt and

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

with an explicit-reasoning prompt that asks the model to reconstruct likely adjacencies before returning labels. We compare both prompts against RANDOM, ALWAYS-NO, and a strong OBSERVED-EDGE baseline that predicts YES only when the candidate edge appeared in the trace.

The result is negative for unobserved topology. Across 1,280 scored query-level examples, direct prompting achieves 72.7% accuracy and explicit reasoning achieves 73.9%, both above the 50% random baseline but below the 82.5% OBSERVED-EDGE baseline. The model is excellent on observed true edges, answering 98.8% correctly under both prompt styles. It fails on the key slice: among true adjacencies that never appeared in the trace, direct prompting recovers only 4/112 (3.6%) and explicit reasoning recovers 0/112. Figure 1 previews the repetition curve, and figure 3 isolates the unobserved-positive failure.

Our contributions are:

- We propose GRAPHTRACE-ADDRESS, a behavioral probe for verbal addressability of in-context graph geometries.
- We conduct a controlled GPT-4.1 evaluation over 2 graph families, 5 context lengths, 8 random seeds, 2 prompt styles, and 1,280 scored adjacency decisions.
- We separate repeated-transition memory from topology inference by scoring observed true edges, low-seen true edges, distance-2 false pairs, and distant false pairs.
- We show that explicit reasoning does not rescue the hard case: it improves some false-neighbor rejection but reduces unobserved true-edge accuracy from 3.6% to 0.0%.

Section 2 positions the study relative to in-context representations and abstract visual reasoning. Section 3 defines the graph tracing benchmark and baselines. Section 4 reports the main results and ablations, and section 5 discusses what this does and does not imply about internal representations.

2 Related Work

In-context graph representations. The closest motivation is the graph tracing setup of Park et al. [2025]. Their work studies whether language-model representations reorganize around context-specified square grids, rings, and other graph structures when a prompt supplies random-walk evidence. Our work keeps the same broad object of study, namely novel graph geometry induced inside a context, but asks a different question. Rather than measuring internal activation geometry or next-token continuation, we ask whether the induced relation is verbally addressable through explicit adjacency queries. This behavioral readout is important because a representation can be structured without being reliably exposed by a prompt.

Abstract visual and spatial reasoning. Several benchmarks show that current multimodal systems struggle with controlled geometric and spatial structure. Bongard-LOGO defines concept-learning problems over procedurally generated 2D shapes with underlying programs Nie et al. [2020]. Shape-Blind shows failures on simple polygon recognition and side counting Rudman et al. [2025]. PuzzleVQA diagnoses abstract visual pattern reasoning by separating perception, induction, and deduction Chia et al. [2024]. Visual Spatial Reasoning tests natural-image spatial relations Liu et al. [2022], while VisuLogic and Spatial-DISE broaden the scope to visual-centric logic and spatial transformations Xu et al. [2025], Huang et al. [2025]. These datasets motivate careful controls for geometry, but most evaluate visual input or multiple-choice classification. We instead use text-only graph traces so that the uncertainty is not image perception but the prompt-accessibility of a newly defined relation.

Prompting, symbolic grounding, and repetition. Work on Bongard-style reasoning finds that direct prompting can underperform structured or descriptive inputs Małkiński et al. [2024]. Symbolic grounding work further shows that models can reason better when abstract visual concepts are represented as programs or language descriptions rather than pixels Vaishnav and Tammet [2026]. These results suggest that failure can come from the interface between representation and query, not only from high-level reasoning. Our study follows the same diagnostic spirit. We compare a terse direct prompt against an explicit-reasoning prompt, but we also add a memorization baseline. This baseline is central: if performance is explained by transitions already observed in context, then the model is addressable as a repeated-edge memory, not as an inferred topology.

3 Methodology

Task definition. We define a hidden graph $G = (V, E)$ with $|V| = 16$. A bijection maps graph nodes to ordinary words, and the model observes a random walk $w_0, w_1, \ldots, w_L$ where each consecutive pair corresponds to an edge in E. After the trace, the model receives adjacency queries (u, v) written as word pairs and must answer `YES` if the corresponding nodes are adjacent and `NO` otherwise. We call an edge observed when the unordered pair appears at least once in the random-walk trace.

Graph families and contexts. We use two graph topologies: GRID4X4, a 4x4 square grid, and RING16, a 16-node cycle. For each topology, we sample 8 random seeds. Each seed fixes the word-to-node assignment, the random walk, and the query bundle. We evaluate context lengths $L \in \{16, 32, 64, 128, 256\}$, roughly corresponding to 1, 2, 4, 8, and 16 visits per node on average. The full design contains 80 graph-context bundles per prompt style.

Queries. Each bundle contains eight adjacency questions drawn from four query types. *Observed true* queries are true graph edges that appeared in the trace. *Low-seen true* queries are true graph edges with low observed counts, including many edges that never appeared. *False distance-2* queries are non-edges whose shortest-path distance is two, and *false distant* queries are non-edges farther away. This split distinguishes exact edge recovery from looser locality.

Prompt styles. We test two prompts. DIRECT asks for JSON-only answers with no visible reasoning. REASONED asks the model to reconstruct likely adjacencies before returning the same scored `YES`/`NO` labels. Both prompts are evaluated with the same ground truth and JSON parser.

Baselines. We compare the model against three baselines. RANDOM predicts a balanced binary decision and has 50% expected accuracy. ALWAYS-NO always predicts `NO`; because the bundles are balanced, it also scores 50% overall, but it exposes class-bias effects. OBSERVED-EDGE predicts `YES` exactly when the queried edge was observed in the trace and `NO` otherwise. This is a strong non-geometric baseline: it rewards repeated transition memory but gives no credit for inferring unobserved edges.

Model and implementation. We call GPT-4.1 through the OpenAI Chat Completions API with temperature 0, API seed 42, and JSON response format. Model availability was checked through the live OpenAI model list on June 1, 2026. The run produced 160 API calls and 1,280 scored query-level examples. Parsing succeeded for every response. Token use was 77,510 prompt tokens and 28,599 completion tokens, for 106,109 total tokens; cost was estimated from the official OpenAI API pricing page OpenAI [2026]. The experiment used Python 3.12.8 with `openai 2.38.0`, `pandas 3.0.3`, `matplotlib 3.10.9`, `scipy 1.17.1`, and `seaborn 0.13.2`. The available machine had four NVIDIA RTX A6000 GPUs, but the experiment was API-bound and did not use GPU acceleration.

Metrics and statistics. The primary metric is query-level accuracy. We report nonparametric 95% bootstrap intervals for table cells and exact binomial tests against the 50% random baseline. We compare paired predictions with exact McNemar tests, including model-vs-OBSERVED-EDGE comparisons within each graph/context cell and a direct-vs-reasoned prompt comparison across matched query bundles. We apply Holm correction for the family of random-baseline tests. Because eight queries are bundled within each trace, query-level examples are not fully independent; we therefore emphasize effect sizes, paired comparisons, and the unobserved-edge slice.

4 Results

Overall accuracy is above random but below memorization. Across all graph types and context lengths, DIRECT reaches 72.7% accuracy and REASONED reaches 73.9%. The deterministic OBSERVED-EDGE baseline reaches 82.5%. Every graph/context cell is above the 50% random baseline after Holm correction, with corrected $p \leq 0.0328$. However, no cell shows a significant model advantage over OBSERVED-EDGE, and OBSERVED-EDGE is significantly better in 10 of 20 model-vs-baseline comparisons.

Repeated edges are verbally addressable. Table 1 shows that longer traces help, especially at $L = 256$, where both prompt styles exceed 84% on both graph families. This gain does not establish topology inference, because the OBSERVED-EDGE baseline also rises with trace length and reaches

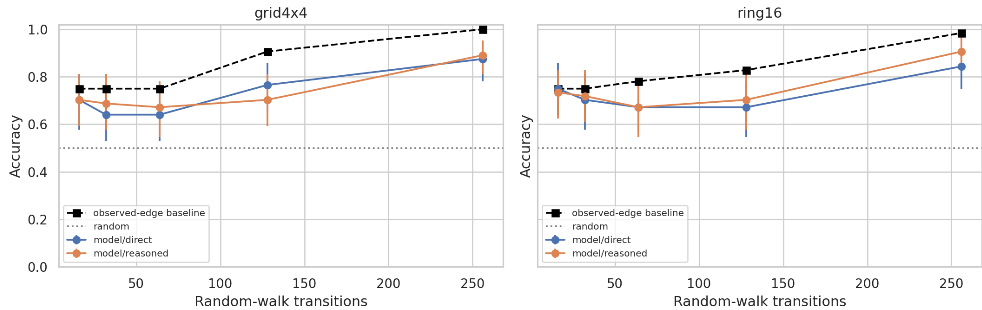

Figure 1: Accuracy by context length. Model accuracy rises at the longest traces, but the curve tracks repeated edge exposure and remains below the observed-edge memorization baseline. Each model point aggregates 64 query-level decisions per graph/context/prompt cell.

Table 1: Accuracy by prompt style, graph topology, and context length. Best result for each graph/-context setting is in **bold**; ties are both bold. The observed-edge baseline is deterministic and shared across prompt styles.

Prompt	Graph	L=16	L=32	L=64	L=128	L=256
DIRECT	GRID4X4	0.703	0.641	0.641	0.766	0.875
DIRECT	RING16	**0.750**	0.703	0.672	0.672	0.844
REASONED	GRID4X4	0.703	0.688	0.672	0.703	0.891
REASONED	RING16	0.734	0.719	0.672	0.703	0.906
OBSERVED-EDGE	GRID4X4	**0.750**	**0.750**	**0.750**	**0.906**	**1.000**
OBSERVED-EDGE	RING16	**0.750**	**0.750**	**0.781**	**0.828**	**0.984**

100.0% on GRID4X4 at $L = 256$ and 98.4% on RING16 at $L = 256$. The strongest behavioral result is more local: observed true edges are almost perfectly addressable. Both prompt styles answer 158/160 observed-true queries correctly, or 98.8%.

The critical slice is unobserved true edges. The low-seen true category in table 2 mixes rare observed edges with unobserved edges. When we isolate true edges with observed count zero, the model nearly always says `NO`. DIRECT recovers only 4/112 unobserved true adjacencies, or 3.6%. REASONED recovers 0/112. In contrast, once a low-seen true edge appears twice, DIRECT answers all 11 such cases correctly; at three or more observations, both prompt styles are perfect on this slice. The failure is therefore not a general inability to answer `YES`; it is a failure to infer missing positive edges from sparse topology.

Explicit reasoning changes errors but does not solve addressability. The explicit-reasoning prompt improves overall accuracy by only 1.25 percentage points, from 72.7% to 73.9%. A paired direct-vs-reasoned McNemar test is not significant ($p = 0.256$). Reasoning helps reject some near false pairs: false distance-2 accuracy increases from 65.0% to 73.8%. But this comes with worse positive recovery in the low-seen slice, which drops from 30.6% to 25.6%, and with complete failure on unobserved true edges.

5 Discussion

The model behaves like a repeated-transition memory. The main result is not that GPT-4.1 cannot use graph traces at all. It clearly can: observed true edges are answered correctly 98.8% of the time, and high-repetition contexts improve accuracy. The result is narrower and more diagnostic. When the queried positive edge has not appeared in the trace, prompt-accessible geometry mostly disappears. This pattern is better explained by local transition memory than by reliable reconstruction of the hidden grid or ring.

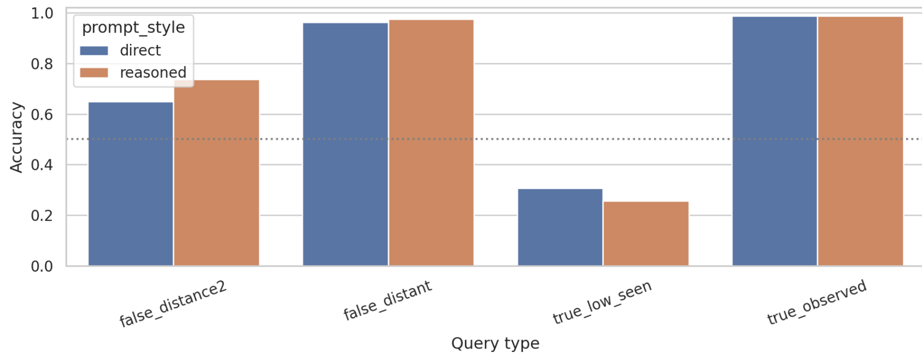

Figure 2: Accuracy by query type. Observed true edges are easy, and distant false pairs are mostly rejected. Low-seen true edges are much harder because this category contains many true edges that never appeared in the trace. False distance-2 pairs are also difficult, indicating that the model sometimes treats graph-near words as adjacent.

Table 2: Accuracy by query type. Best result in each column is in **bold**. Low-seen true edges include unobserved positives and rare observed positives; the unobserved-only slice is analyzed separately in the text and in figure 3.

Prompt or baseline	Observed true	Low-seen true	False distance-2	False distant
DIRECT	0.988	**0.306**	0.650	0.963
REASONED	0.988	0.256	0.738	0.975
OBSERVED-EDGE	**1.000**	0.300	**1.000**	**1.000**

Near-neighbor errors suggest vague locality without exact edges. False distant pairs are easy, with 96.3% accuracy under DIRECT and 97.5% under REASONED. False distance-2 pairs are harder, with 65.0% and 73.8% accuracy respectively. This contrast suggests that the model may acquire a coarse notion that some words live in the same local region of the trace, while still failing to resolve exact adjacency. Such a representation would be useful for broad locality judgments but insufficient for graph addressability.

Reasoning is not a reliable interface here. The explicit-reasoning prompt was meant to test whether terse JSON-only prompting hides an available structure. It did not. Reasoning slightly improves rejection of false distance-2 pairs but does not improve the central positive case, and it reduces unobserved true-edge accuracy to zero. This does not prove that internal geometric representations are absent. It shows that the tested reasoning prompt does not make those representations behaviorally available as exact adjacency answers.

Limitations. This is a behavioral API study, not an activation analysis. It cannot decide whether GPT-4.1 internally formed a geometric representation that the prompt failed to access, or whether no such representation formed. We test one model family, while the motivating representation study analyzes other models. The random walks are unlabeled, so they identify adjacency but not absolute directions such as north or east. The bundled query design also means that the effective sample size is closer to the 80 graph-context bundles per prompt style than to 640 fully independent queries per prompt style.

Broader implications. These results argue for separating in-context representation learning from prompt addressability. A model may store repeated local relations well enough to answer many questions, yet still fail when asked about relations implied by the whole structure. For applications that define temporary maps, workflows, databases, or symbolic worlds inside a prompt, this distinction matters: repeated evidence can look like understanding unless evaluation isolates unobserved but entailed relations.

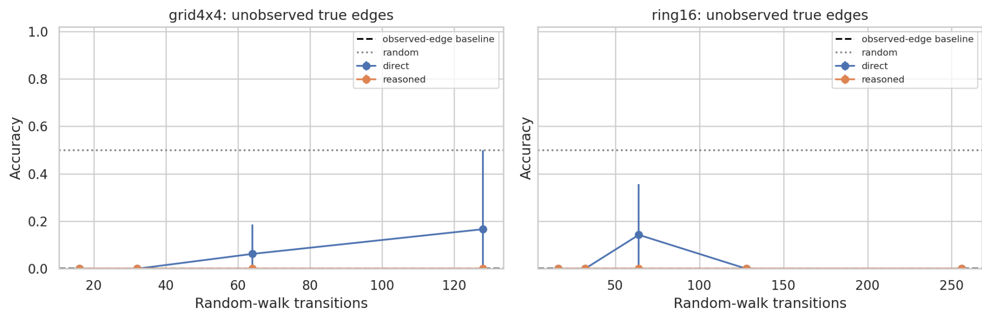

Figure 3: Accuracy on true graph edges that never appeared in the prompt trace. The model rarely recovers these missing positives, even though they are valid adjacencies in the hidden graph. Explicit reasoning does not improve this slice.

6 Conclusion

We tested whether newly specified graph geometries are verbally addressable from random-walk context. In a controlled benchmark with hidden 4x4 grids and 16-node rings, GPT-4.1 answered repeated edges reliably but did not recover unobserved true adjacencies. Direct prompting achieved 72.7% overall accuracy, explicit reasoning achieved 73.9%, and both remained below the 82.5% observed-edge memorization baseline. On the decisive unobserved-positive slice, direct prompting scored 4/112 and explicit reasoning scored 0/112.

The key takeaway is simple: in this setting, repeated novel geometries are addressable mainly when the relevant transition has already appeared in the prompt. Future work should test GPT-5-class and open-weight models, add directed move labels for coordinate probes, compare against explicit graph-reconstruction algorithms, and combine behavioral prompts with activation analyses on the same graph traces.

References

Yew Ken Chia, Vernon Toh Yan Han, Deepanway Ghosal, Lidong Bing, and Soujanya Poria. Puzzlevqa: Diagnosing multimodal reasoning challenges of language models with abstract visual patterns, 2024. URL `https://arxiv.org/abs/2403.13315`.

Xinmiao Huang, Qisong He, Zhenglin Huang, Boxuan Wang, Zhuoyun Li, Guangliang Cheng, Yi Dong, and Xiaowei Huang. Spatial-dise: A unified benchmark for evaluating spatial reasoning in vision-language models, 2025. URL `https://arxiv.org/abs/2510.13394`.

Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning, 2022. URL `https://arxiv.org/abs/2205.00363`.

Mikołaj Małkiński, Szymon Pawlonka, and Jacek Mańdziuk. Reasoning limitations of multimodal large language models. a case study of bongard problems, 2024. URL `https://arxiv.org/abs/2411.01173`.

Weili Nie, Zhiding Yu, Lei Mao, Ankit B. Patel, Yuke Zhu, and Animashree Anandkumar. Bongard-logo: A new benchmark for human-level concept learning and reasoning. In *Advances in Neural Information Processing Systems*, pages 16480–16492, 2020. URL `https://arxiv.org/abs/2010.00763`.

OpenAI. Openai api pricing. `https://platform.openai.com/docs/pricing`, 2026. Accessed June 1, 2026.

Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, and Hidenori Tanaka. Iclr: In-context learning of representations, 2025. URL `https://arxiv.org/abs/2501.00070`.

William Rudman, Michal Golovanevsky, Amir Bar, Vedant Palit, Yann LeCun, Carsten Eickhoff, and Ritambhara Singh. Forgotten polygons: Multimodal large language models are shape-blind, 2025. URL https://arxiv.org/abs/2502.15969.

Mohit Vaishnav and Tanel Tammet. Symbolic grounding reveals representational bottlenecks in abstract visual reasoning, 2026. URL https://arxiv.org/abs/2604.21346.

Weiye Xu, Jiahao Wang, Weiyun Wang, Zhe Chen, Wengang Zhou, Aijun Yang, Lewei Lu, Houqiang Li, Xiaohua Wang, Xizhou Zhu, Wenhai Wang, Jifeng Dai, and Jinguo Zhu. Visulogic: A benchmark for evaluating visual reasoning in multi-modal large language models, 2025. URL https://arxiv.org/abs/2504.15279.